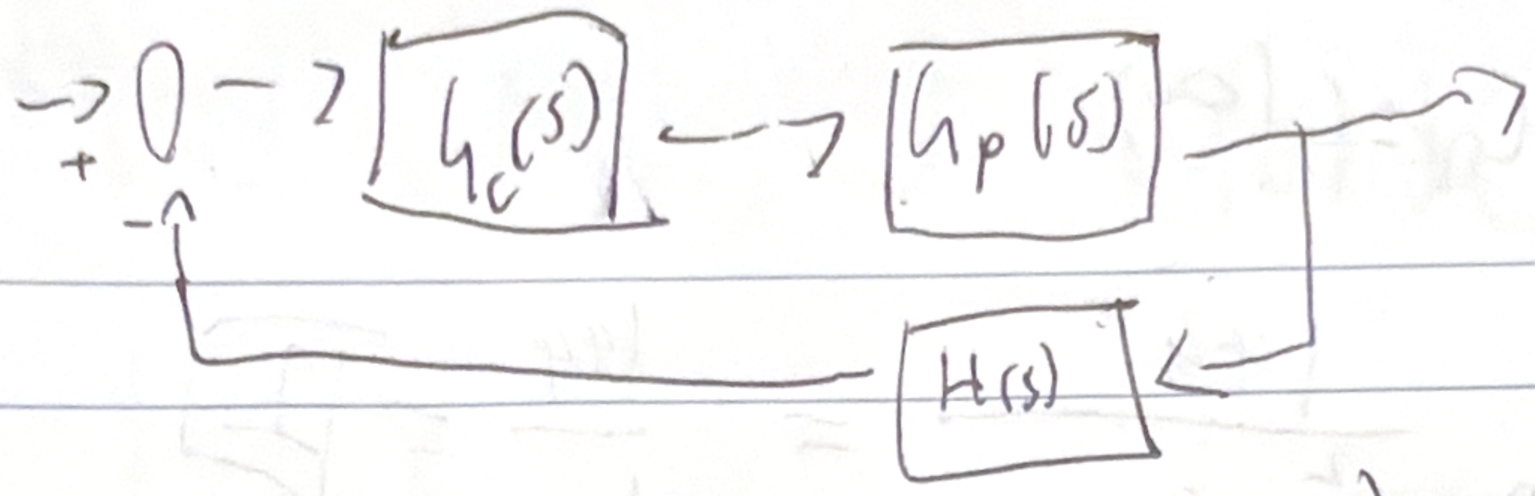
(2)



$G_c(s) = K$ $\quad G_p(s) = \dfrac{1}{(s+2)(s-10)(s-12)}$

$H(s) = 1$

a) $G(s) = G_c(s)\,G_p(s)\,H(s) = \dfrac{K}{(s+2)(s-10)(s-12)}$

$p = -2, 10, 12$ $\quad$ **$G(s)$ has 3 poles**

b) $T(s) = \dfrac{G(s)}{1+G(s)} = \dfrac{K/(s+2)(s-10)(s-12)}{1 + K/(s+2)(s-10)(s-12)}$

$= \dfrac{K}{(s+2)(s-10)(s-12)+K} = K/s^3 - 20s^2 + 76s + 240 + K$

$K=1$: $\dfrac{1}{s^3 - 12s^2 + 241}$ $\quad s \approx -2, 10, 12$

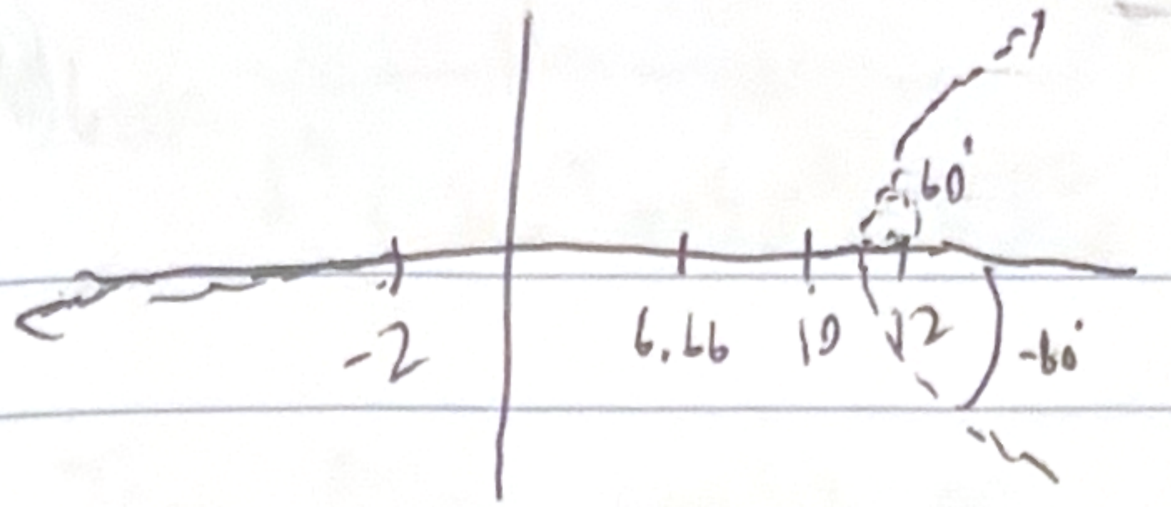
still has 2 right hand poles
even if they aren't exactly the same

c) $K=2$: $\dfrac{2}{s^3 - 12s^2 + 242}$ $\quad s \approx -2, 10, 12$

Once again, still has 2 RHP poles
even if they are not exactly
the same.

$$C = \frac{-2+10+12}{3} = 6.66$$

f)



$$\theta_{A_0} = \frac{2(0)+1}{3} 180° = 60°$$

$$\theta_{A_1} = \frac{2(1)+1}{3} 180° = 180°$$

$$\theta_{A_2} = \frac{2(2)+1}{3} 180° = 300°$$

e) is the same?

g) ~~$G_c(s) =$~~

yes you can change $G_c(s)$ to make the sy stem stable.

h)

$$\boxed{G_c(s) = K(s-10)(s-12)}$$

2.

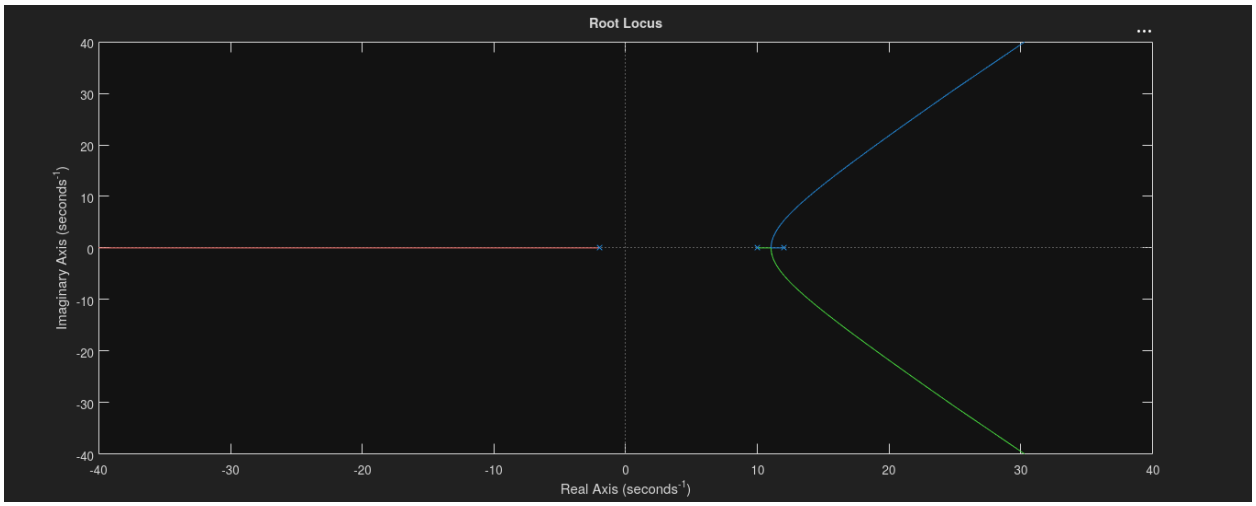
f)

```
>> sys = tf(NUM,DEN)

sys =

             1
  -------------------------
  s^3 - 20 s^2 + 76 s + 240

Continuous-time transfer function.
Model Properties
>> rlocus(sys)
>>
```



i)

```
>> sys = tf([1],[1 2])

sys =

    1
  -----
  s + 2

Continuous-time transfer function.
Model Properties
>> rlocus(sys)
>>
```

